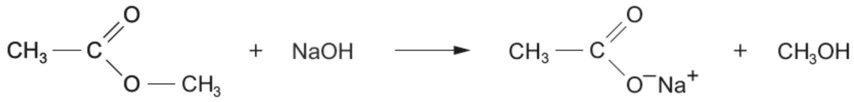
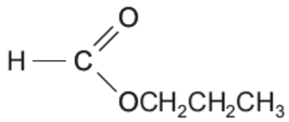
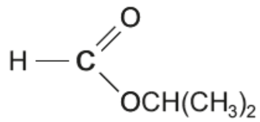
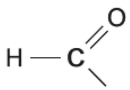
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			**Indicative content** • percentage of chlorine in this compound is 32.1% which is consistent with that of 2,4-D • colourless gas with $NaHCO_3$ indicates compound is a carboxylic acid • n(NaOH) is 1.85×10^{-3} mol • mole ratio is 1:1 therefore only one COOH group • number of moles of compound in 25.0 cm^3 is also 1.85×10^{-3} mol therefore 1.85×10^{-2} in 250 cm^3 • M_r of compound is 221 which is consistent with that of 2,4-D test results consistent with structure of 2,4-D • no chloride ions on reflux with NaOH therefore chlorine atoms are bonded directly to the ring, not in alkyl side-chains • no white precipitate with aqueous bromine therefore not a phenol • aqueous bromine not decolourised therefore no C=C double bonds spectral data consistent with that of 2,4-D • ^{1}H NMR spectrum of this compound would show three peaks; aromatic (area of 3), —CH_2— (area 2), O—H (area 1) • ^{13}C NMR of this compound would show eight peaks as it has 8 different carbon environments • take its melting temperature and compare to a book value • mix a sample with some actual 2,4-D, and take its melting temperature – see if the value is unchanged	2	2	2	6	2	4

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
				5-6 marks All information has been used and interpreted correctly, good account of melting temperature determination *The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary is used accurately throughout.* **3-4 marks** Most of the information has been used and interpreted correctly *The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.* **1-2 marks** Some of the information has been used correctly *The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give an answer worthy of credit.*						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)				1		1		
		(ii)	I	88		1		1	1	
			II	 or  (1) ester must contain  group for it to reduce Tollens reagent (1)		1	1	2		
		(iii)	I	60		1		1	1	
			II	if relative molecular masses are different, the number of moles present in the chromatogram is not directly related to the mass of each component			1	1		
		(iv)		**A** has weaker intermolecular forces therefore (less energy is needed to overcome these forces and) it has a lower boiling temperature			1	1		
		(v)		2-methylbut-1-ene			1	1		
				Question 9 total	**2**	**6**	**6**	**14**	**4**	**4**